

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

CE-5005(3)-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Environmental Engineering

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.
iii) Assume any missing data.

1. Determine the future population of a satellite town for the year 2021 by Arithmetic Increase method and compare the result by using Geometric increase method. 14

Year	1951	1961	1971	1981	1991
Population (in thousand)	93	111	132	161	197

2. Design a sewer to serve a population of 36,000; the daily per capita water supply allowance being 135 liters, of which 80 percent finds its way into the sewer. The slope available for the sewer to be laid is 1 in 625 and the sewer should be designed to carry four times the dry weather flow when running full. What would be the velocity of flow in the sewer when running full? 14
Assume $n = 0.012$ in Manning's formula

3. Write note on following with neat sketches if required: 14
a) Water Distribution System
b) Manhole and its classification

CE-5005(3)-CBGS

PTO

[2]

4. a) Discuss how a slow sand filter differs from a rapid sand filter? 7
b) State and explain the basic principles of operation involved in the treatment of waste water by oxidation pond. 7
5. a) Describe methods of purification of water. 7
b) What is coagulation? Discuss the mechanism of coagulation. 7
6. Explain the following terms: 14
 - i) Suspended and dissolved particles
 - ii) Chlorination
 - iii) Sludge digestion
 - iv) Types of server
7. a) Explain the following: 14
 - i) Disinfection and sterilisation, minor methods of disinfection
 - ii) Septic tank and Imhoff tank
8. Write a short notes on: 14
 - i) Socket and spigot joints
 - ii) Skimming Tank
 - iii) Sedimentation with coagulation

CE-5005(3)-CBGS